آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Pacing Guide for Teachers

PHYSICS

Grade XII

Theory

FOR ACADEMIC YEAR 2026 AND ONWARDS

Number of weeks: 28

Number of periods per week: 5

Key Textbook: Model Textbook of Physics 12, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Teacher Developer(s) and their Institution(s):

Muhammad Ali Rana - Ghulaman-e-Abbas School, Karachi

Israr-ul-Haq - Ghulaman-e-Abbas School, Karachi

Saib Ahmed Khan - Govt. Degree Science and Commerce College, Orangi Township, Karachi

Qasim Abbas - Aga Khan Higher Secondary School, Kharadar, Karachi

Muhammad Affan Khan - Aga Khan Higher Secondary School, Karimabad, Karachi

Syeda Shazia Ahmed - Habib Girl's Higher Secondary School, Karachi

Muhammad Salman - Habib Public High School, Karachi

Topic	Total Periods
11. Gravitation	11

Sub-Topic	Range of SLOs	Periods (40 mins)
11.1 Newton's Law of Gravitation	11.1.1 – 11.1.2	02
11.2 Gravitational Field Strength	11.2.1 – 11.2.2	02
11.3 Gravitational Force and Motion of Satellites	11.3.1 – 11.3.3	02
11.4 Variation in value of 'g'	11.4.1 – 11.4.2	02
11.5 Gravitational Potential	11.5.1 – 11.5.2	03

FOR ACADEMIC YEAR 2026 AND ONWARDS

Web Resources

https://www.youtube.com/watch?v=0ewWkfBA3vw?t=210
http://www.enchantedlearning.com/subjects/astronomy/planets/

Suggested Activities and/ or Formative Assessment

Activity 1:

Provide students with the formula for calculating the force of gravity [$F=G(m_1m_2/r^2)$] and ask them to calculate the force between different objects (such as the sun and the earth). This activity can help students develop problem-solving skills while learning about the mathematical aspects of Newton's Law of Gravitation.

Learning Resources

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics%20Final.pdf

Further Resources

For additional resources related to teaching, learning and formative assessments, please refer to **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
12. Electrostatics	20

Sub-Topic	Range of SLOs	Periods (40 mins)
12.1 Charge	12.1.1 – 12.1.2	01
12.2 Coulomb's Law	12.2.1 – 12.2.6	02
12.3 Electric Field and Electric Field Intensity	12.3.1 – 12.3.5	03
12.4 Electric Flux	12.4.1	02
12.5 Gauss's Law	12.5.1 – 12.5.2	03
12.6 Electric Potential	12.6.1 – 12.6.6	03
12.7 Capacitor	12.7.1 – 12.7.5	03
12.8 Energy Stored in a Capacitor	12.8.1 – 12.8.4	03

FOR ACADEMIC YEAR 2026 AND ONWARDS

Learning Resources

- Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.
- Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.
- Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).
- Sears, Zamansky and Young Addison, College Physics (2011). New York: Wesley Publication Co.

Web Resources

https://www.youtube.com/watch?v=f_MZNsEqyQw

https://ophysics.com/em5.html

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
13. Electric Current	22

Sub-Topic	Range of SLOs	Periods (40 mins)
13.1 Current Electricity	13.1.1 – 13.1.3	02
13.2 Resistance and Resistivity	13.2.1 – 13.2.6	02
13.3 Electromotive Force and Potential Difference	13.3.1 – 13.3.5	04
13.4 Electric Power	13.4.1 – 13.4.3	03
13.5 Kirchhoff's Laws	13.5.1 – 13.5.5	04
13.6 Potential Divider	13.6.1 – 13.6.3	03
13.7 D.C. Circuits	13.7.1 – 13.7.4	04

FOR ACADEMIC YEAR 2026 AND ONWARDS

Learning Resources

- Sears, Zamansky and Young Addison, College Physics. New York: Wesley Publication Co.
- David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, Cambridge International AS and *A Level Physics Coursebook Second Edition (2014)*: Cambridge University Press.
- Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).

Suggested Activities and/ or Formative Assessment

Activity 1:

Ask students to use the given circuit simulator and make circuits and determine different quantities, e.g., current, voltage, resistance, etc.

https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
14. Electromagnetism	12

Sub-Topic	Range of SLOs	Periods (40 mins)
14.1 Magnetic Field by a Current Carrying Conductor	14.1.1 – 14.1.7	03
14.2 Magnetic Force	14.2.1 – 14.2.5	03
14.3 Magnetic Torque	14.3.1 – 14.3.2	03
14.4 Practical Applications of Electromagnetism	14.4.1 – 14.4.5	03

Learning Resource

- Halliday and Resnick, Fundamentals of Physics: Walker Wiley India Pvt. Ltd. (Volume 2)

Suggested Activities and/ or Formative Assessment

Activity 1:

The given simulation can be used in the classroom.

https://javalab.org/en/category/electricity_en/electromagnetism_en/

AKU-EB Teaching and Learning Resource Guide:

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
15. Electromagnetic Induction and Alternating Current	18

Sub-Topic	Range of SLOs	Periods (40 mins)
15.1 Law of Electromagnetic Induction	15.1.1 – 15.1.4	02
15.2 Seismometer	15.2.1 – 15.2.2	02
15.3 Inductance	15.3.1 – 15.3.2	02
15.4 Alternating Current (A.C.) as Sinusoidal Wave	15.4.1 – 15.4.6	03
15.5 A.C. Circuits	15.5.1 – 15.5.6	03
15.6 Semiconductor Devices	15.6.1 – 15.6.4	03
15.7 Rectification and Smoothing	15.7.1 – 15.7.4	03

FOR ACADEMIC YEAR 2026 AND ONWARDS

Web Resources

Working Principle of DC Generator | [Electric Machine #1] - YouTube

https://ophysics.com/em11.html

https://ophysics.com/em10.html

Suggested Activities and/ or Formative Assessment

Activity 1:

For basic concept of induction: https://ophysics.com/em11.html

Activity 2:

For the introduction of DC generator: https://ophysics.com/em10.html

Learning Resources

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resources%20Guide%202025%20HSSC%20Physics%20Final.pdf

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
16. Quantum Mechanics	20

Sub-Topic	Range of SLOs	Periods (40 mins)
16.1 Einstein Special Theory of Relativity	16.1.1 – 16.1.5	04
16.2 Quantum Theory	16.2.1 – 16.2.3	03
16.3 Photoelectric Effect	16.3.1 – 16.3.5	04
16.4 Compton's Effect	16.4.1 – 16.4.2	02
16.5 Wave-Particle Duality	16.5.1 – 16.5.6	04
16.6 Atomic Structure and Heisenberg Uncertainty Principle	16.6.1 – 16.6.7	03

FOR ACADEMIC YEAR 2026 AND ONWARDS

Web Resource

https://www.youtube.com/watch?v=30KfPtHec4s

Learning Resources

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
17. Nuclear Physics	12

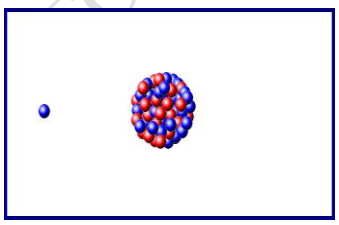
Sub-Topic	Range of SLOs	Periods (40 mins)
17.1 Structure of Nucleus	17.1.1 – 17.1.2	01
17.2 Mass Defect and Binding Energy	17.2.1 – 17.2.4	02
17.3 Radioactivity	17.3.1 – 17.3.6	02
17.4 Radioactive Decay Law	17.4.1 – 17.4.8	02
17.5 Nuclear Reactions	17.5.1 – 17.5.2	02
17.6 Nuclear Reactor	17.6.1 – 17.6.3	03

FOR ACADEMIC YEAR 2026 AND ONWARDS

Suggested Activities and/ or Formative Assessment

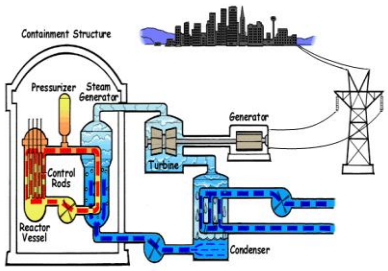
Activity 1:

Show the animation and ask students to discuss how it is happening?



Activity 2:

Show the animation of the nuclear reactor and ask to discuss about the procedure.



Learning Resources

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ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
18. Particle Physics	08

Sub-Topic	Range of SLOs	Periods (40 mins)
18.1 Basic Forces of Nature	18.1.1	01
18.2 Fundamental Particles	18.2.1 – 18.2.5	03
18.3 Quarks And Anti-quarks	18.3.1 – 18.3.6	02
18.4 Particle Accelerator	18.4.1	02

Web Resource

https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/University_Physics_III_-_Optics_and_Modern_Physics_(OpenStax)/11%3A_Particle_Physics_and_Cosmology/11.02%3A_Introduction_to_Particle_Physics

https://www.quantamagazine.org/a-new-map-of-the-standard-model-of-particle-physics-20201022/

Suggested Activities and/ or Formative Assessment

Teacher can ask students to sit in groups and create mind map of “Particles in Physics” using https://coggle.it. If they do not have their laptops in classroom then this activity can also be assigned as their homework.

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Learning Resources

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
19. Medical and Space Physics	13

Sub-Topic	Range of SLOs	Periods (40 mins)
19.1 Production and Use Of Ultrasound	19.1.1 – 19.1.7	02
19.2 Production and Use of X-rays	19.2.1 – 19.2.4	02
19.3 PET Scanning	19.3.1 – 19.3.2	02
19.4 Luminosity and Inverse Square law	19.4.1 – 19.4.4	02
19.5 Stellar Radii	19.5.1 – 19.5.4	02
19.6 Hubble Law	19.6.1 – 19.6.3	03

Web Resources

https://physics.uiowa.edu/research/space-physics

https://www.iaea.org/resources/hhc/medical-physics

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FOR ACADEMIC YEAR 2026 AND ONWARDS

Topic	Total Periods
20. Nature of Science	04

Sub-Topic	Range of SLOs	Periods (40 mins)
20.1 Debates about Aesthetic in Physics	20.1.1 – 20.1.3	02
20.2 Thought Experiments	20.2.1 – 20.2.2	02

Web Resource

https://www.youtube.com/watch?v=xFKgIOX8IRE

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Note: This teacher-led pacing guide has been developed by educators from AKU-EB affiliated schools to support fellow teachers teaching the AKU-EB syllabi by

- ensuring smooth transition of a school's academic year.
- ensuring curricular continuity in schools.
- predicting the time and pace of syllabi implementation.

This document also contains **suggested activities and/ or formative assessments** that may enhance the learning experience. Please note that these activities are meant to serve as suggestions. As educators, you have the flexibility and autonomy to adapt and modify them to best suit the needs of your students and the dynamics of your classroom.

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